

Relay — Data Model

Status: Draft for build · **Owner:** Systems/Backend · **Last updated:** June 2026

PostgreSQL 15 + pgvector. Moss holds the production retrieval index; Postgres is the system of record (and the pgvector fallback). Single-org assumption for the hackathon — `organization_id` is included for forward-compatibility but defaulted.

Entity overview

```
organizations 1——* documents 1——* chunks
organizations 1——* users
sessions 1——* utterances
sessions 1——* cards
cards *——* chunks      (via card_sources)
customers 1——* memories
customers 1——* sessions (optional, Desk)
sessions 1——1 leads    (Intake)
```

Tables

organizations

column	type	notes
id	uuid PK	org_...
name	text	
created_at	timestampz	default now()

users

column	type	notes
id	uuid PK	usr_...
organization_id	uuid FK	
name	text	
role	text	

documents

column	type	notes
id	uuid PK	doc_...
organization_id	uuid FK	
title	text	

column	type	notes
source_type	text	pdf docx txt
status	text	processing ready failed
tags	text[]	
chunk_count	int	
created_at	timestampz	

chunks

column	type	notes
id	uuid PK	chk_...
document_id	uuid FK	
ordinal	int	position in doc
text	text	chunk content
embedding	vector(1024)	pgvector fallback index
moss_ref	text	id/handle in the Moss index
metadata	jsonb	page, section, table flags

Index: ivfflat (embedding vector_cosine_ops) for fallback. Primary path queries Moss by moss_ref.

sessions

column	type	notes
id	uuid PK	ses_...
organization_id	uuid FK	
mode	text	live desk intake
customer_id	uuid FK null	Desk/Intake
livekit_room	text null	
status	text	active ended
started_at / ended_at	timestampz	

utterances

column	type	notes
id	uuid PK	utt_...
session_id	uuid FK	
speaker	text	e.g. rep prospect customer

column	type	notes
text	text	final transcript
ts	timestampz	

cards

column	type	notes
id	uuid PK	card_...
session_id	uuid FK	
mode	text	
answer	text	grounded, synthesized
trigger_text	text	what prompted it
latency_ms	int	trigger → render
created_at	timestampz	

card_sources (join: card ↔ chunk)

column	type	notes
card_id	uuid FK	
chunk_id	uuid FK	
score	float	retrieval score
PK	(card_id, chunk_id)	

customers

column	type	notes
id	uuid PK	CUS_...
organization_id	uuid FK	
name / company / email	text	

memories (cross-session, Desk/Intake)

column	type	notes
id	uuid PK	mem_...
customer_id	uuid FK	
kind	text	fact summary preference
text	text	
embedding	vector(1024)	semantic recall

column	type	notes
source_session_id	uuid FK null	
created_at	timestampz	

leads (Intake)

column	type	notes
id	uuid PK	lead_...
session_id	uuid FK	
name / company / email	text	
qualifiers	jsonb	budget, timeline, need
score	int	0–100 (ICP fit)
status	text	hot warm cold
routed_to	text null	channel/owner
created_at	timestampz	

Retrieval data flow

1. **Ingest:** documents row (status=processing) → Unsiloed parse → chunks rows → embed → write to Moss (moss_ref) + pgvector (embedding) → status=ready.
2. **Query:** text → Moss top-k → resolve moss_ref → chunks → orchestrator → cards + card_sources.
3. **Fallback:** Moss unavailable → pgvector cosine search over chunks.embedding.

Related docs

API_SPEC.md · TECHNICAL_DESIGN.md